

The Renewal Threshold: Predicting PG Resident Renewal Intention Using Behavioral and Service-Experience Analytics

Paying Guest (PG) Accommodation Services

Analytical dataset: supplied 10,000-row synthetic survey-like CSV

Report basis: analysis.ipynb in this project folder

Scope note: no raw survey export or proposal file was available; the analytical file is treated as synthetic and is not described as 10,000 real respondents.

Individual academic case study

1. Problem Statement and Objectives

Business problem. Paying Guest (PG) accommodation operators need to retain residents while maintaining service quality and reasonable costs. Amenities and security are visible, conventional service dimensions, while complaint handling, complaint volume, deposit-refund transparency and perceived rent fairness capture ongoing service experience. This study asks whether the latter block shows stronger association and predictive information for stated renewal intention in the supplied analytical file.

Research question. Do behavioral/service-experience factors—complaint resolution, complaint count, deposit refund transparency and rent fairness—predict renewal intention more strongly than amenities and security ratings? Overall satisfaction is treated separately as a descriptive and outcome-like measure, and is excluded from the primary comparison.

Objectives. (1) Describe renewal responses and the distribution of the principal experience variables. (2) Compare amenity-only, service-only and combined feature sets using statistical association tests and Logistic Regression/Random Forest cross-validation. (3) Form practical recommendations that match the strength and limits of the evidence.

Decision relevance. If operational service measures add useful predictive information, managers could prioritize complaint tracking, rent-value communication and transparent deposit policies alongside amenity maintenance. The analysis is exploratory and cannot establish that changing a feature will cause renewal.

2. Data Collection and Dataset Description

Dataset used. The primary file, *data/synthetic_survey_10000.csv*, contains 10,000 rows and 12 survey-like columns: occupation; monthly rent; stay duration; amenities rating; security rating; complaint resolution; complaint count; deposit refundability; rent fairness; satisfaction; renewal intention; and biggest reason.

Provenance and collection. The faculty accepted the supplied 10,000-row synthetic CSV as the approved analytical dataset for this case study. The original Google Forms export, approved proposal and a verifiable generation script were not present in the supplied project materials. Faculty acceptance establishes this file as the approved analytical dataset; it does not make the synthetic records real respondents. The collection source, recruitment procedure, consent process and source-to-synthetic fidelity cannot be independently audited from this file alone.

Response distribution. The file contains 4,706 Yes (47.06%), 4,194 Maybe (41.94%) and 1,100 No (11.00%). For binary modeling, only definite Yes and No responses are included (n=5,806; Yes rate 81.05%); Maybe (4,194) is not recoded as No and is excluded from model fitting.

Composition. There are 9,145 working-professional rows and 855 student rows. Rent ranges from 5,000 to 18,000, stay duration from 1 to 36 months, amenities and security from 1 to 5, complaint count from 0 to 6, and satisfaction from 2 to 5. These are file descriptions, not verified population estimates.

Privacy. The supplied file contains no name or timestamp columns. No real-person identifiers were added. Since the source survey was not supplied, anonymization of the original collection cannot be audited.

3. Data Preparation and Exploratory Analysis

Inspection. The input has 10,000 rows, 12 columns, zero missing cells and 69 exact duplicate rows. Duplicate synthetic rows were retained because the supplied dataset is being analyzed as delivered and the records cannot be checked against original timestamps or identities. They are a potential sign of repeated/sampled patterns, not evidence of duplicate real respondents. No names or timestamps are available to inspect.

Variable handling. Analysis-friendly fields are derived in memory while the original CSV remains unchanged. Amenities/security and satisfaction remain on their 1–5 scales; rent, stay months and complaint count remain numeric. Deposit responses are ordered as straightforward=0, not sure=1, disputes common=2. Rent fairness is coded underpriced=0, fair=1, overpriced=2. Complaint resolution is encoded same day=0, few days=1, over a week=2, never resolved=3; no complaints raised is a separate model category (4), not treated as unresolved.

Consistency checks. Complaint-count/resolution combinations were flagged descriptively: 2,273 rows show zero complaints with a complaint-resolution response, and 2,033 show a positive complaint count alongside “no complaints raised.” These logical conflicts were retained rather than silently deleted. The resolution variable has 2,941 “no complaints raised” rows, which are kept as their own group.

Unavailable preprocessing checks. Rent and stay duration are already numeric in the provided file; no raw text formats remain to test the prescribed “k” rent parsing or ambiguous duration-unit assumptions. A sensitivity comparison of alternative duration interpretations is not possible without raw source responses and generation provenance. The generated range is retained without claiming source-level correction.

Leakage control. “Biggest reason” is excluded from all predictive features because it may directly express the answer rationale. Satisfaction is excluded from the main comparison and examined separately as a proxy/leakage sensitivity. The questionnaire has no unexpected-charges field; no such feature is created.

3. Data Preparation and Exploratory Analysis (continued)

Descriptive patterns. Average amenities rating is 3.99 (median 4), average security rating 4.12 (median 4), average satisfaction 4.56 (median 5), and average complaint count 1.14 (median 1). These high ratings and response distributions describe the synthetic file only.

Renewal by amenities/security. Definite Yes proportions increase across amenities ratings overall: 32.4% at rating 2, 38.2% at rating 3, 47.6% at rating 4, and 54.9% at rating 5. The one rating-1 record is too sparse for interpretation. For security, Yes ranges from 32.2% at rating 2 to 52.2% at rating 5, with just one observation at rating 1.

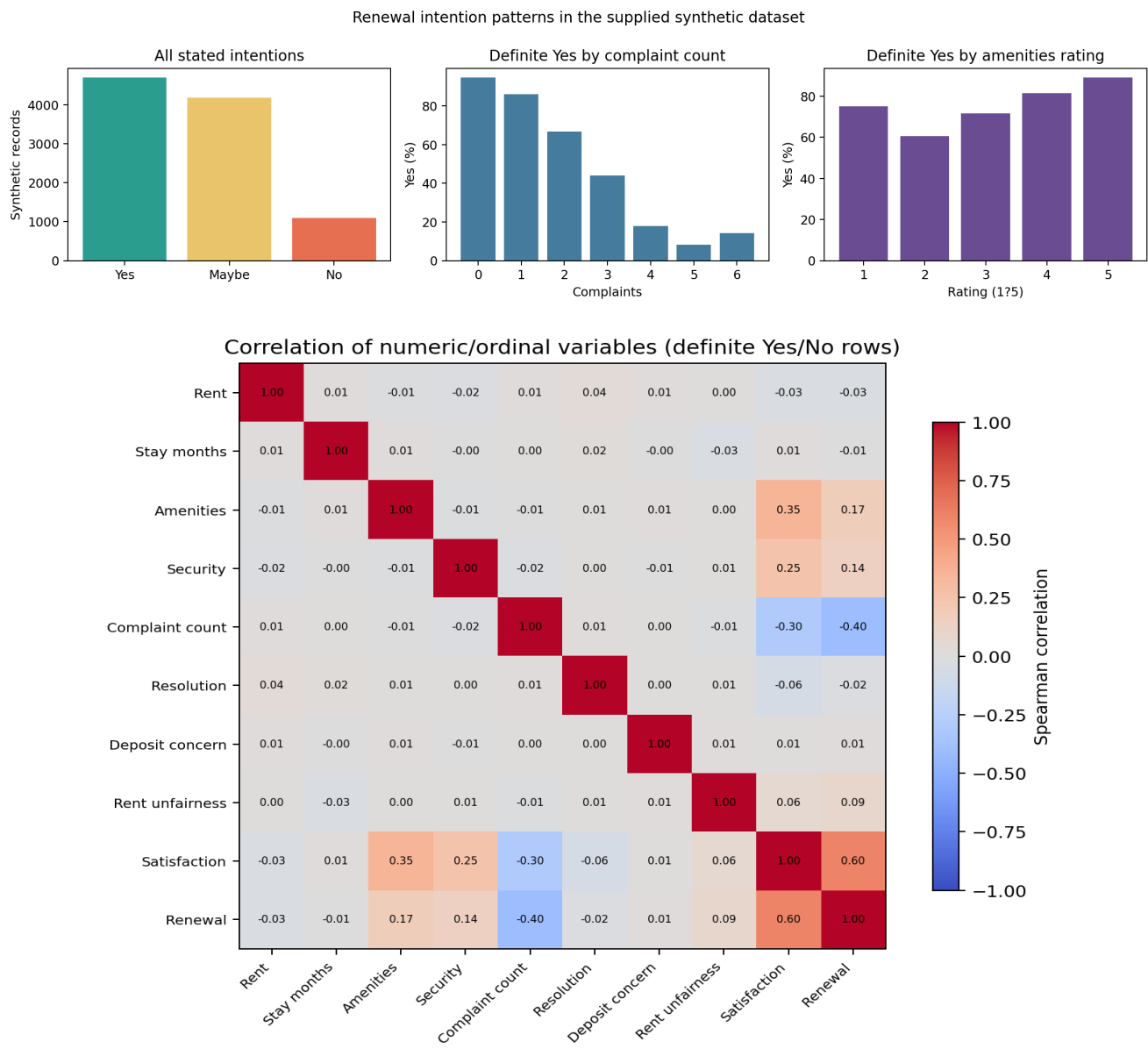
Renewal by service measures. Yes is 64.3% for complaint count 0, 48.8% for count 1, 32.9% for count 2, 20.8% for count 3, and 10.6% for count 4. Categories above 4 have small counts. Yes is 24.5% for “underpriced,” 49.5% for “fair,” and 20.2% for “overpriced.” Deposit categories have similar Yes proportions (46.8%–49.7%). Resolution categories also show similar proportions (41.3%–48.2%).

Satisfaction. Definite Yes is 61.7% at satisfaction 5, 25.6% at 4, 6.1% at 3 and 0% at 2. This very strong relationship supports keeping satisfaction outside the core driver comparison because it is outcome-like and may closely proxy intention.

Interpretation. Complaint count and rent fairness show notable gradients; deposit transparency and resolution do not show similarly clear gradients in this file. Category rates are descriptive, not causal. The pattern may reflect the synthetic generation rules, and repeated rows may affect apparent precision.

3. Data Preparation and Exploratory Analysis: Visual Summary

Figure 1 summarizes the three intention classes and the observed definite-Yes gradients by complaint count and amenities rating. Figure 2 shows Spearman associations among numeric and ordinal variables in the definite Yes/No subset. Encodings are descriptive; correlations do not imply causation.



4. Analytics Method and Implementation

Why these methods. Cross-tabulations and conditional renewal rates provide interpretable descriptive comparisons. Mann–Whitney U tests compare ordinal/numeric distributions between definite Yes and No, with rank-biserial correlation as effect size. Chi-square and Cramér’s V summarize categorical association. Logistic Regression provides an interpretable linear baseline; Random Forest captures nonlinear splits. Neither model is treated as a causal model.

Feature blocks. Amenity-only = amenities and security. Service-only = complaint resolution, complaint count, deposit refund concern and rent unfairness. Combined = all six core variables. “No complaint raised” is encoded distinctly from never resolved. Satisfaction is added only in a separate experiment; “biggest reason” is excluded.

Validation. Models use 5-fold stratified cross-validation repeated 10 times (50 held-out folds) with seed 42. Median imputation and standardization for Logistic Regression are fitted inside each training fold. Metrics are ROC-AUC, average precision, balanced accuracy and F1. Reported 2.5th–97.5th percentiles are empirical fold quantiles, not formal population confidence intervals.

Class imbalance. Among the binary subset, 81.1% are Yes and 18.9% No. A majority-class classifier therefore reaches 81.1% accuracy while offering no class discrimination; balanced accuracy is 0.50 for this baseline. ROC-AUC, balanced accuracy and average precision are reported to give a fuller view.

Important caveat. Repeated CV partitions the supplied synthetic records; it does not validate against newly collected residents. Where records share a common synthetic generation process, folds may contain very similar cases. Strong scores can therefore reflect generator logic.

4. Analytics Method and Implementation: Statistical Results

Association tests. On the 5,806 definite Yes/No records, amenities differ by intention (Mann–Whitney $U=3,205,818$; $p<.001$; rank-biserial $r=0.239$), and security differs ($U=3,080,671$; $p<.001$; $r=0.190$). Complaint count has a larger inverse association with Yes ordering ($U=1,144,582$; $p<.001$; $r=-0.558$). Rent unfairness has a smaller association ($U=2,752,602$; $p<.001$; $r=0.064$).

Weak/no detected associations. Resolution category is not statistically distinguishable in this test ($U=2,522,000$; $p=.150$; $r=-0.026$); deposit refund concern is not ($U=2,614,292$; $p=.506$; $r=0.010$). Occupation is also weakly associated ($\chi^2=0.287$; $p=.592$; Cramér's $V=0.008$). Satisfaction is strongly associated ($U=4,396,682$; $p<.001$; $r=0.699$), reinforcing its separate treatment.

How to read the tests. Because the file is synthetic, p-values describe differences within this constructed dataset and are not evidence from 5,806 independently sampled real residents. Small effects such as rent-fairness rank-biserial correlation may be statistically detectable at this scale but modest in magnitude. Complaint-count effect size is comparatively larger.

Concordance with descriptive rates. Complaint count's inverse relationship is evident in the rate table. Deposit and resolution rates vary little between categories. These descriptive and inferential summaries point in the same direction but inherit the same synthetic-data limitations.

5. Comparison with State-of-the-Art Methods

Three recent studies address renewal/revisit intention in adjacent property and hospitality settings. Their populations, outcomes and methods differ; scores or statistical significance are not directly rankable against this case. Where accessible abstracts do not report a metric, the comparison says so rather than inferring one.

Published study / year	Dataset	Method used	Evaluation metric	Key result	Comparison with this work
In (2024), commercial property renewal	Commercial real-estate tenant context; sample size/details not in accessible record	Management reliability, adequacy and tenant empathy examined against perceived value and renewal intention	Hypothesis/influence testing; numeric metric not reported in record	All three service-quality dimensions positively related to perceived value and renewal intention	Shares property renewal/service focus; our predictors are PG survey features and predictive CV. Our file emphasizes complaint volume; their abstract emphasizes management quality.
Marandi, Tasavori & Najmi (2023 online; journal 2024)	About 100,000 TripAdvisor reviews for 100 global hotels	Machine learning/text analysis; feature extraction and segmentation	Text feature/segment analysis; predictive score not reported in repository abstract	71 features; room, staff, food and accessibility salient; 15 customer segments	Hospitality revisit outcome and much larger real review corpus. Our structured synthetic survey tests predefined blocks, not textual feature discovery.
Park & Lee (2025)	Online self-report survey of single-person households in Busan; n/metrics not given in accessible abstract	Hypothesis testing of location factors and mediation by residential satisfaction	Significance/mediation tests; metric values not in abstract	Location factors related to renewal; satisfaction mediated some relationships	Relevant residential lease-renewal context; ours concerns PG service experience and synthetic data. Their geographic/sample limitations resemble the need for cautious generalization.

Source note. In (2024): KCI record and DOI 10.47116/apjcri.2024.08.16. Marandi et al.: University of Essex repository record and DOI 10.1108/IJCHM-06-2022-0719. Park & Lee: journal record and DOI 10.6106/KJCEM.2025.26.5.053. Where an abstract did not disclose dataset size or metric values, the table says so.

6. Results, Business Insights and Recommendations

Block comparison. In repeated CV, service-only outperforms amenity-only on mean ROC-AUC for both models: Logistic Regression 0.787 vs 0.660; Random Forest 0.806 vs 0.654. The combined block is highest: 0.831 for Logistic Regression and 0.842 for Random Forest. Random Forest balanced accuracy is 0.743 for service-only and 0.762 combined, compared with 0.609 amenity-only. This synthetic-file comparison suggests service features carry more predictive information than amenity/security alone, with additional information from combining blocks.

Baseline and precision. Majority-class accuracy is 0.811; this superficially high baseline underscores why raw accuracy is unsuitable. Combined Random Forest mean average precision is 0.950, while baseline prevalence is 0.811. These metrics are internal repeated-CV summaries on synthetic records and should not be treated as deployment performance.

Feature importance. On one stratified test split, permuting complaint count reduces combined Random Forest ROC-AUC by 0.223 on average; next are amenities (0.048), rent unfairness (0.037), security (0.029), resolution (0.009) and deposit concern (0.005). Importance is model- and split-dependent and may reflect programmed relationships; it is not a causal effect.

Satisfaction proxy experiment. Adding satisfaction raises mean ROC-AUC from 0.831 to 0.911 in Logistic Regression and from 0.842 to 0.913 in Random Forest. This notable increase demonstrates why satisfaction is analyzed separately: it may encode an outcome-like summary or be closely tied to renewal intention. It is not used to answer the primary block comparison.

Recommendation 1 — Monitor complaints. Track complaint counts and resolution status consistently, investigate repeat issues, and review whether service response workflows address resident concerns. Complaint volume shows the strongest relationship in this dataset; validate against actual renewal outcomes before allocating resources based on a score.

Recommendation 2 — Make rent value and deposit terms clear. Communicate rent inclusions and refund conditions in writing; use complaints and fairness feedback to prioritize review. Deposit transparency showed little separation here, so this recommendation is a low-risk service practice rather than a proven renewal driver.

Recommendation 3 — Preserve core amenities and security. Continue maintenance and safety monitoring. Both ratings associate with intention and add predictive value when combined with service features, even though their block alone performs less strongly in this file.

7. Conclusion and References

Conclusion. On the supplied synthetic file, behavioral/service features—especially complaint count—show stronger predictive association than amenity/security ratings alone. Service-only cross-validation exceeds amenity-only, and combining both blocks performs best. The evidence therefore supports the research question within this generated dataset, but cannot establish an equivalent relationship among real PG residents or show that an intervention would cause higher renewal.

Limitations. The original collected sample, raw survey, proposal and reproducible synthesis provenance were not supplied; thus actual original sample size, non-renewer count, respondent composition and cleaning decisions cannot be verified. Synthetic expansion adds no real-world observations. The 10,000 rows include 69 exact duplicates and contradictory complaint/resolution combinations. The Yes/No modeling subset excludes 4,194 Maybe rows. Measures are self-report-like and cross-sectional; sampling and geography are unverified. The file is dominated by working professionals (91.45%). There is no unexpected-charges variable. Ambiguous raw rent/duration entries cannot be audited from already numeric values. Ordinal coding, class imbalance within Yes/No, possible generator rule leakage and unstable feature importance limit conclusions. No causal inference is possible.

Next step. Replicate on a new, ethically collected and anonymized sample with documented recruitment and synthesis lineage, retain Maybe as a distinct outcome, and measure actual renewal behavior over time. Report performance on untouched real cases before operational use.

References

- In, S. (2024). A study on the impact of commercial real estate management service quality on tenants' intention to renew their contracts. *Asia-Pacific Journal of Convergent Research Interchange*, 10(8), 199–210. <https://doi.org/10.47116/apicri.2024.08.16>
- Marandi, A. A., Tasavori, M., & Najmi, M. (2023; journal issue 2024). New insights into hotel customers' revisiting intentions, based on big data. *International Journal of Contemporary Hospitality Management*, 36(1), 292–311. <https://doi.org/10.1108/IJCHM-06-2022-0719>
- Park, Y.-J., & Lee, Y.-D. (2025). The effect of housing location factors on lease renewal intention among single-person households. *Korean Journal of Construction Engineering and Management*, 26(5), 53–65. <https://doi.org/10.6106/KJCEM.2025.26.5.053>
- Data source: supplied file `pg\_synthetic\_10000.csv`, copied to this project as `data/synthetic\_survey\_10000.csv`. No independent collection website or raw questionnaire export was available for citation.